Test 2 (CH. 5 – CH. 7)

Name_____ Date_____

Multiple choice. (50 pts. 2 pts each). Circle the correct answer.

1. Determine the molarity of a solution formed by dissolving 97.7 g LiBr in enough water to yield 750.0 mL of solution.
   A) 1.50 M
   B) 1.18 M
   C) 0.130 M
   D) 0.768 M
   E) 2.30 M

2. Give the complete ionic equation for the reaction (if any) that occurs when aqueous solutions of sodium sulfide and iron(II) nitrate are mixed.
   A) $2\ Na^{+}(aq) + S^{2-}(aq) + Fe^{2+}(aq) + 2\ NO_3^{-}(aq) \rightarrow Fe^{2+}(aq) + S^{2-}(aq) + 2\ Na^{+}(aq) + 2\ NO_3^{-}(aq)$
   B) $2\ Na^{+}(aq) + S^{2-}(aq) + Fe^{2+}(aq) + 2\ NO_3^{-}(aq) \rightarrow FeS(s) + 2\ NaNO_3(s)$
   C) $2\ Na^{+}(aq) + S^{2-}(aq) + Fe^{2+}(aq) + 2\ NO_3^{-}(aq) \rightarrow Fe^{2+}(aq) + S^{2-}(aq) + 2\ NaNO_3(s)$
   D) $2\ Na^{+}(aq) + S^{2-}(aq) + Fe^{2+}(aq) + 2\ NO_3^{-}(aq) \rightarrow FeS(s) + 2\ Na^{+}(aq) + 2\ NO_3^{-}(aq)$
   E) No reaction occurs.

3. Give the net ionic equation for the reaction (if any) that occurs when aqueous solutions of sodium sulfide and iron(II) nitrate are mixed.
   A) $Na^{+}(aq) + NO_3^{-}(aq) \rightarrow NaNO_3(s)$
   B) $S^{2-}(aq) + Fe^{2+}(aq) \rightarrow FeS(s)$
   C) $2\ Na^{+}(aq) + S^{2-}(aq) + Fe^{2+}(aq) + 2\ NO_3^{-}(aq) \rightarrow Fe^{2+}(aq) + S^{2-}(aq) + 2\ NaNO_3(s)$
   D) $2\ Na^{+}(aq) + S^{2-}(aq) + Fe^{2+}(aq) + 2\ NO_3^{-}(aq) \rightarrow FeS(s) + 2\ Na^{+}(aq) + 2\ NO_3^{-}(aq)$
   E) No reaction occurs.

4. Identify the spectator ions for the reaction that occurs when aqueous solutions of sodium sulfide and iron(II) nitrate are mixed.
   A) Fe^{2+} and NO_3^{-}
   B) Na^{+} and Fe^{2+}
   C) Na^{+} and NO_3^{-}
   D) S^{2-} and Fe^{2+}
   E) No spectator ions are present.

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5. What element is undergoing reduction (if any) in the following reaction?

$$CH_4(g) + 2\ O_2(g) \rightarrow CO_2(g) + 2\ H_2O(g)$$

A) O
B) H
C) C
D) both C and H
E) None of the elements is undergoing oxidation.

6. A student prepared a stock solution by dissolving 20.0 g of KOH in enough water to make 150. mL of solution. She then took 15.0 mL of the stock solution and diluted it with enough water to make water to make 65.0 mL of a final solution.
A) 0.548 M
B) 0.713 M
C) 1.40 M
D) 1.82 M

7. In an acid-base neutralization reaction, 43.74 mL of 0.500 M potassium hydroxide reacts with 50.00 mL of sulfuric acid solution. What is the concentration of the H_2SO_4 solution?

$$H_2SO_4 + 2KOH \rightarrow K_2SO_4 + 2H_2O$$

A) 0.219 M
B) 0.437 M
C) 0.875 M
D) 1.14 M

8. Which one of the following compounds is insoluble in water?
A) K Br
B) KNO_3
C) Pb Cl_2
D) Li_2CO_3

9. Give the temperature and pressure at STP.
A) 0°C and 760 mm Hg
B) 100K and 760 torr
C) 25°C and 30.00 in Hg
D) 273 K and 1 torr Hg
E) 0°C and 1 mm Hg

Test 2 (CH. 5 – CH. 7)

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10. A large balloon is initially filled to a volume of 25.0 L at 353 K and a pressure of 2575 mm Hg. What volume of gas will the balloon contain at 1.35 atm and 253 K?

A) 22.2 L
B) 87.5 L
C) 11.4 L
D) 45.0 L
E) 58.6 L

11. Convert 1.25 atm to torr.

A) 760 torr
B) 775 torr
C) 950 torr
D) 1000 torr
E) 2400 torr

12. A mixture of 0.220 moles CO, 0.350 moles SO_2 and 0.640 moles He has a total pressure of 2.95 atm. What is the partial pressure of SO_2?

A) 1.17 atm
B) 0.853 atm
C) 1.03 atm
D) 0.969 atm
E) 0.649 atm

13. A gas occupies 3.67 L at 2.22 atm. What is the volume at 4.21 atm?

A) 6.96 L
B) 2.55 L
C) 1.94 L
D) 22.1 L
E) 0.517 L

14. What pressure will 14.0 g of CO exert in a 3.5 L container at 75°C?

A) 4.1 atm
B) 5.0 atm
C) 6.4 atm
D) 1.1 atm
E) 2.3 atm

Test 2 (CH. 5 – CH. 7)

Name_____ Date_____

15. An unknown gas has a mass of 1.56 g. Its volume is 0.50 L at a temperature of 315 K and a pressure of 1.14 atm.

A) Br_2

B) HCl

C) H_2O

D) Cl_2

E) HBr

16. To what volume will a 2.33 L sample of gas expand if it is heated from 20.0°C to 200.0°C?

A) 3.76 L

B) 24.3 L

C) 0.445 L

D) 1.44 L

E) 0.693 L

17. Determine the specific heat capacity of an alloy that requires 69.3 kJ to raise the temperature of 150.0 g alloy from 298 K to 398 K.

A) 2.20 J/g°C

B) 2.80 J/g°C

C) 4.62 J/g°C

D) 1.79 J/g°C

E) 7.00 J/g°C

18. Identify what a bomb calorimeter measures.

A) measures ΔH for aqueous solutions

B) measures ΔE for combustion reactions

C) measures ΔH for oxidation solutions

D) measures ΔT for hydrolysis solutions

E) measures ΔE for oxidation reactions

19. For a particular process that is carried out at constant pressure, q = 165 kJ and w = -55 kJ. Therefore,

A) ΔE = 110 kJ and ΔH = 165 kJ.

B) ΔE = 165 kJ and ΔH = 110 kJ.

C) ΔE = 165 kJ and ΔH = 220 kJ.

D) ΔE = 220 kJ and ΔH = 165 kJ.

E) ΔE = 15 kJ and ΔH = 45 kJ.

Test 2 (CH. 5 – CH. 7)

Name_____ Date_____

20. Calculate ΔHrxn for the reaction: N_2O (g) + NO_2 (g) → 3 NO (g)

Use the following reactions and given ΔH's:

2 NO (g) + O_2 (g) → 2 NO_2 (g)	ΔH = −113.1 kJ
N_2 (g) + O_2 (g) → 2 NO (g)	ΔH = +182.6 kJ
2 N_2O (g) → 2 N_2 (g) + O_2 (g)	ΔH = −163.2 kJ

A) 44.45 kJ
B) −25.05 kJ
C) −93.70 kJ
D) 157.6 kJ
E) 122.8 kJ

21. Suppose that 16 g of aluminum is initially at 23.0°C. What is the final temperature of the aluminum upon absorbing 3.25 kJ of heat? The specific heat capacity of aluminum is 0.903 J/g • °C.
A) 23.0°C
B) 102°C
C) 225°C
D) 248°C
E) 332°C

22. Use the ΔH°f information provided to calculate ΔH°rxn for the following:

Cr_2O_3 (s) + 3 CO (g) → 2 Cr (s) + 3 CO_2 (g)

ΔH°f (kJ/mol)	
Cr_2O_3 (s)	−1139.6
CO (g)	−110.5
Cr (s)	0.0
CO_2 (g)	−393.5

A) −265.6 kJ
B) 856.6 kJ
C) 290.7 kJ
D) −1643.6 kJ
E) −2650.7 kJ

Test 2 (CH. 5 – CH. 7)

Name_____ Date_____

23. Hydrogen peroxide decomposes to water and oxygen at constant pressure by the following reaction:

$$2\ H_2O_2\ (l) \rightarrow 2\ H_2O(l) + O_2\ (g) \qquad \Delta H = -196\ kJ$$

Calculate the value of q (kJ) in this exothermic reaction when 4.00 g of hydrogen peroxide decomposes at constant pressure.

A) -23.1 kJ
B) -11.5 kJ
C) -0.0217 kJ
D) 1.44 kJ
E) -2.31 × 104 kJ

24. When 0.250 mol of $CH_4(g)$ reacts with excess $Cl_2(g)$ at constant pressure according to the chemical equation shown below, 177 kJ of heat are released. Calculate the value of ΔH for this reaction, as written.

$$2\ CH_4(g) + 3\ Cl_2(g) \rightarrow 2\ CHCl_3(l) + 3\ H_2(g) \qquad \Delta H = ?$$

A) -1420 kJ
B) -708 kJ
C) +708 kJ
D) +1420 kJ
E) -2122 kJ

25. A balloon is inflated from 0.0100 L to 0.400 L against an external pressure of 10.00 atm. How much work is done in joules? 101.3 J = 1 L × atm

A) -39.5 J
B) 675 J
C) +0.395 J
D) 395 J
E) -395 J

EQUATIONS:

$$pV = nRT \qquad P_1V_1 = P_2V_2. \qquad \frac{V_1}{N_1} = \frac{V_2}{N_2}$$

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}. \qquad \frac{V_1}{T_1} = \frac{V_2}{T_2} \qquad \frac{P_1}{T_1} = \frac{P_2}{T_2}$$

R = 0.08206 L˙atm / mol˙K

$$KE = \tfrac{1}{2}mv^2 \qquad \Delta E = q + w \qquad q = C \times \Delta T$$

$$q = m \times C_s \times \Delta T \qquad w = -P_{ext}\Delta V \qquad q_{cal} = C_{cal} \times \Delta T = -q_{rxn}$$

$$C_{S,H_2O} = 4.184\ J/g°C \qquad H = E + PV$$

Test 2 (CH. 5 – CH. 7)

Name_____ Date_____

SOLUBILITY RULES:

Ion	Solubility	Exceptions
NO_3^-	soluble	none
ClO_4^-	soluble	none
Cl^-	soluble	except Ag^+, Hg_2^{2+}, *Pb^{2+}
I^-	soluble	except Ag^+, Hg_2^{2+}, Pb^{2+}
SO_4^{2-}	soluble	except Ca^{2+}, Ba^{2+}, Sr^{2+}, Hg^{2+}, Pb^{2+}, Ag^+
CO_3^{2-}	insoluble	except Group IA and NH_4^+
PO_4^{3-}	insoluble	except Group IA and NH_4^+
-OH	insoluble	except Group IA, *Ca^{2+}, Ba^{2+}, Sr^{2+}
S^{2-}	insoluble	except Group IA, IIA and NH_4^+
Na^+	soluble	none
NH_4^+	soluble	none
K^+	soluble	none

*slightly soluble